

JEE(Main + Advanced) : ENTHUSIAST COURSE : [SCORE-I (ALL STAR BATCH)]

ANSWER KEY

PART-1 : PHYSICS

SECTION-I	Q.	1	2	3	4	5	6	7	8	9	10
	A.	C	A	D	B	C	D	A	D	D	B
SECTION-II	Q.	11	12	13	14	15	16	17	18	19	20
	A.	D	B	C	D	D	A	A	C	D	D

PART-2 : CHEMISTRY

SECTION-I	Q.	1	2	3	4	5	6	7	8	9	10
	A.	B	D	D	C	B	A	B	A	D	A
SECTION-II	Q.	11	12	13	14	15	16	17	18	19	20
	A.	C	C	D	C	B	C	A	A	C	D

PART-3 : MATHEMATICS

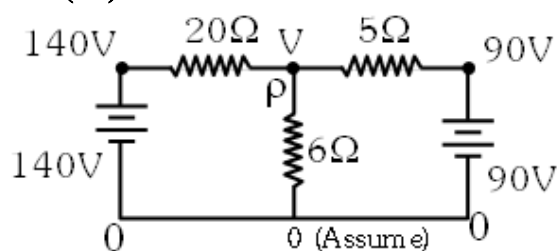
SECTION-I	Q.	1	2	3	4	5	6	7	8	9	10
	A.	B	B	A	C	C	A	B	C	C	A
SECTION-II	Q.	11	12	13	14	15	16	17	18	19	20
	A.	B	D	D	D	B	A	B	B	B	A

HINT – SHEET

PART-1 : PHYSICS

SECTION-I

1. Ans (C)



Applying KCL at point P,

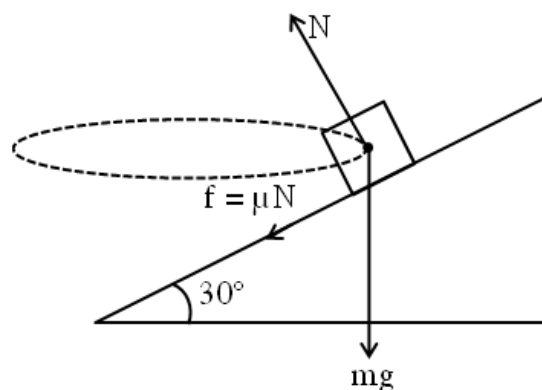
$$\frac{V - 0}{6} + \frac{V - 90}{5} + \frac{V - 140}{20} = 0$$

$$\Rightarrow 10V + 12V - 1080 + 3V - 420 = 0$$

$$\Rightarrow V = 60$$

$$\therefore \text{current in } 6\Omega = \frac{V - 0}{6} = 10A$$

2. Ans (A)

At $v_{\max}$, f will be limiting in nature.

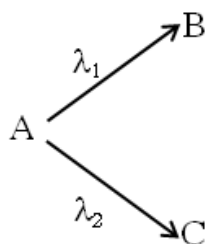
∴ Balancing force in vertical direction,

$$N \cos 30^\circ - mg - \mu N \cos 60^\circ = 0$$

$$\Rightarrow N [\cos 30^\circ - \mu \cos 60^\circ] = mg$$

$$\therefore N = \frac{800 \times 10}{(0.87 - 0.1)} \approx 10.2 \times 10^3 \text{ kg m/s}^2$$

3. Ans (D)



Given $\lambda_1 = \frac{\ln 2}{700}/\text{year}$, $\lambda_2 = \frac{\ln 2}{1400}/\text{year}$

$$\therefore \lambda_{\text{net}} = \lambda_1 + \lambda_2 = \ln 2 \left[\frac{1}{700} + \frac{1}{1400} \right]$$

$$= \frac{3 \ln 2}{1400}/\text{year}$$

Now, Let initial no. of radioactive nuclei be N_0 .

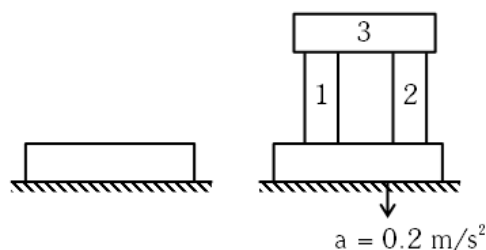
$$\therefore \frac{N_0}{3} = N_0 e^{-\lambda_{\text{net}} t}$$

$$\Rightarrow \ln \frac{1}{3} = -\lambda_{\text{net}} t$$

$$\Rightarrow 1.1 = \frac{3 \times 0.693}{1400} t$$

$$\Rightarrow t \approx 740 \text{ years}$$

4. Ans (B)



Writing force equation in vertical direction

$$Mg - N = Ma$$

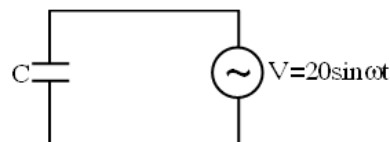
$$\Rightarrow 70g - N = 70 \times 0.2$$

$$\Rightarrow N = 70 [g - 0.2] = 70 \times 9.8$$

$$\therefore N = 686 \text{ Newton}$$

Note : Since there is no compressive normal from the sides, hence friction will not act

5. Ans (C)



From the given information,

$$C = \frac{\epsilon_0 A}{d} = \frac{\epsilon_0 \times 1}{2 \times 10^{-3}} \text{ F}$$

$$\therefore X_C = \frac{1}{\omega C} = \frac{2 \times 10^{-3}}{2 \times 50\pi \times \epsilon_0} = \frac{2 \times 10^{-3}}{25 \times 4\pi \epsilon_0} \Omega$$

$$\therefore X_C = \frac{2 \times 10^{-3}}{25} \times 9 \times 10^9 = \frac{18}{25} \times 10^6 \Omega$$

$$\therefore i_0 = \frac{V_0}{X_C} = \frac{20 \times 25}{18} \times 10^{-6} \text{ A}$$

$$= 27.47 \mu\text{A}.$$

The value of amplitude of displacement current will be same as value of amplitude of conventional current.

6. Ans (D)

$$\frac{\mu_2}{v} - \frac{\mu_1}{u} = \frac{\mu_2 - \mu_1}{R}$$

$$\frac{1.4}{v} - \frac{1.25}{-40} = \frac{1.4 - 1.25}{-25}$$

$$\frac{1.4}{v} = -\frac{0.15}{25} - \frac{1.25}{40}$$

$$v = -37.58 \text{ cm}$$

7. Ans (A)

At neutral point $g = 0$ so graph (C) is correct

8. Ans (D)

$$\Delta Q = \Delta U + \Delta W$$

$$\text{Here } \Delta W = 0$$

$$\Delta Q = \Delta U = nC_V \Delta T$$

$$\Delta Q = 4 \times \frac{5R}{2} (50) = 500R$$

9. Ans (D)

$$\vec{A} \cdot \vec{B} = |\vec{A} \times \vec{B}|$$

$$AB \cos \theta = AB \sin \theta \Rightarrow \theta = 45^\circ$$

$$|\vec{A} - \vec{B}| = \sqrt{A^2 + B^2 - 2AB \cos 45^\circ}$$

$$= \sqrt{A^2 + B^2 - \sqrt{2}AB}$$

10. Ans (B)

$$r = \frac{mv}{qB} = \frac{\sqrt{2mk}}{qB}$$

$$\frac{r_d}{r_a} = \sqrt{\frac{m_d}{m_a} \frac{q_a}{q_d}} = \sqrt{\frac{2}{4}} \left(\frac{2}{1} \right) = \sqrt{2}$$

11. **Ans (D)**

Energy of γ ray $[E_\gamma] = h\nu$

Momentum of γ ray $[P_\gamma] = \frac{h}{\lambda} = \frac{h\nu}{C}$

Total momentum is conserved.

$$\vec{P}_\gamma + \vec{P}_{Nu} = 0$$

Where $\vec{P}_{Nu}$ = Momentum of decayed nuclei

$$\Rightarrow P_\gamma = P_{Nu}$$

$$\Rightarrow \frac{h\nu}{C} = P_{Nu}$$

$\Rightarrow$ K.E. of nuclei

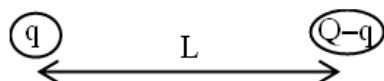
$$\Rightarrow \frac{1}{2}Mv^2 = \frac{(P_{Nu})^2}{2M} = \frac{1}{2M} \left[\frac{h\nu}{C} \right]^2$$

Loss in internal energy = $E_\gamma + K.E_{Nu}$

$$= h\nu + \frac{1}{2M} \left[\frac{h\nu}{C} \right]^2$$

$$= h\nu \left[1 + \frac{h\nu}{2MC^2} \right]$$

12. **Ans (B)**



$$F_q = \frac{kq(Q-q)}{L^2} = \frac{k}{L^2}(qQ - q^2)$$

$\frac{dF}{dq} = 0$ when force is maximum

$$\frac{dF}{dq} = \frac{k}{L^2}[Q - 2q] = 0$$

$$\Rightarrow Q - 2q = 0 \Rightarrow Q = 2q$$

13. **Ans (C)**

$$I = \vec{J} \cdot \vec{A} = JA \cos(\theta)$$

$$5 = J \left(\frac{4}{100} \right) \times \cos(60)$$

$$J = 5 \times 50 = 250 \text{ A/m}^2$$

$$\text{Now, } \vec{E} = \rho \cdot \vec{J}$$

$$= 44 \times 10^{-8} \times 250 = 11 \times 10^{-5} \text{ V/m}$$

14. **Ans (D)**

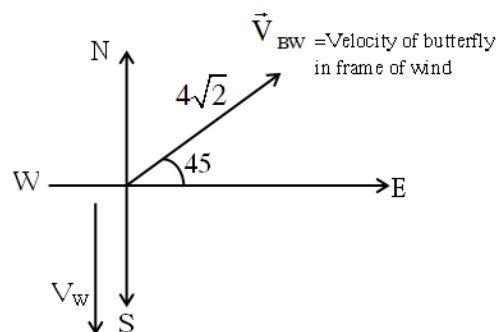
$$V_{RMS} = \sqrt{\frac{3RT}{M}}$$

$$m_A < m_B < m_C$$

$$\Rightarrow V_A > V_B > V_C$$

$$\Rightarrow \frac{1}{V_A} < \frac{1}{V_B} < \frac{1}{V_C}$$

15. **Ans (D)**



$$\vec{V}_{BW} = 4\sqrt{2} \cos 45^\circ \hat{i} + 4\sqrt{2} \sin 45^\circ \hat{j}$$

$$= 4\hat{i} + 4\hat{j}$$

$$\vec{V}_W = -\hat{j}$$

$$\vec{V}_B = \vec{V}_{BW} + \vec{V}_W = 4\hat{i} + 3\hat{j}$$

$$\vec{S}_B = \vec{V}_B \times t = (4\hat{i} + 3\hat{j}) \times 3 = 12\hat{i} + 9\hat{j}$$

$$|\vec{S}_B| = \sqrt{(12)^2 + (9)^2} = 15\text{m}$$

16. **Ans (A)**

$$Y = \frac{FL}{A\Delta L}$$

$$\Rightarrow Y = \frac{T_1 \ell_0}{A(\ell_1 - \ell_0)} = \frac{T_2 \ell_0}{A(\ell_2 - \ell_0)}$$

$$1 = \frac{T_1(\ell_2 - \ell_0)}{T_2(\ell_1 - \ell_0)}$$

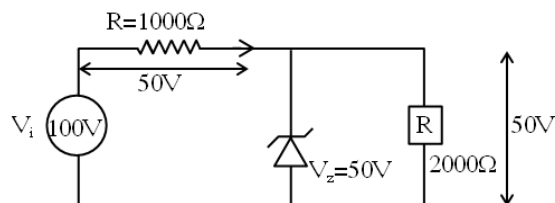
$$T_2 \ell_1 - T_2 \ell_0 = T_1 \ell_2 - T_1 \ell_0$$

$$(T_1 - T_2)\ell_0 = T_1 \ell_2 - T_2 \ell_1$$

$$\ell_0 = \left(\frac{T_1 \ell_2 - T_2 \ell_1}{T_1 - T_2} \right)$$

17. **Ans (A)**

$$I = \frac{50}{1000} = 50\text{mA}$$



$$I = \frac{50}{2000} = 25\text{mA}$$

$$I_Z = I_{1000} - I_{2000}$$

$$= 50 - 25 = 25 \text{ mA}$$

18. Ans (C)

As rod moves in field area increases upto $x = b$ then field is absent and again flux is generated on return journey from $x = b$ to $x = 0$. Thus plot A for flux.

$$\Rightarrow e = -\frac{d\phi}{dt} \Rightarrow \text{curve B for emf}$$

$\Rightarrow$ Power dissipated = $vi \Rightarrow$ curve C for power dissipated

19. Ans (D)

$$S = \alpha^2 \beta \ln \left(\frac{\mu KR}{J\beta^2} + 3 \right)$$

$$S = \frac{Q}{T} = \text{joule/k}$$

$$[\alpha^2 \beta] = \text{Joule/k}$$

$$PV = nRT \quad \left[\frac{\mu KR}{J\beta^2} \right] = 1$$

$$R = \frac{\text{Joule}}{K}$$

$$\Rightarrow R = \frac{\text{Joule}}{K}, K = \frac{\text{Joule}}{R}$$

$$\Rightarrow \beta = \left(\frac{\text{Joule}}{K} \right)$$

$$\alpha^2 \beta = \left(\frac{\text{Joule}}{K} \right)$$

$$\Rightarrow \alpha = \text{dimensionless}$$

20. Ans (D)

$$1.51 \text{ ————— } n = 3$$

$$3.4 \text{ ————— } n =$$

$$13.6 \text{ ————— } n = 1$$

$$3 \rightarrow 2 \Rightarrow 1.89 \text{ eV}$$

$$5 \times 10^{-4} \text{ T} \quad r = 7 \text{ mm}$$

$$r = \frac{mv}{qB} \Rightarrow mv = qrB$$

$$\Rightarrow E = \frac{p^2}{2m} = \frac{(qRB)^2}{2m}$$

$$= \frac{(1.6 \times 10^{-19} \times 7 \times 10^{-3} \times 5 \times 10^{-4})^2}{2 \times 9.1 \times 10^{-31} \text{ Joule}}$$

$$= \frac{3136 \times 10^{-52}}{18.2 \times 10^{-31} \times 1.6 \times 10^{-19}} \text{ eV}$$

$$= 1.077 \text{ eV}$$

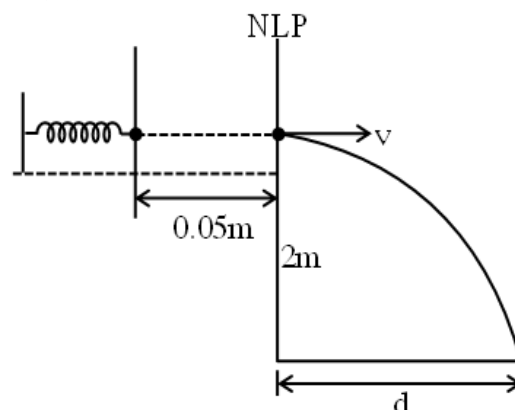
We know work function = energy incident - $(KE)_{\text{electron}}$

$$\phi = 1.89 - 1.077 = 0.813 \text{ eV}$$

PART-1 : PHYSICS

SECTION-II

1. Ans (1)



$$\frac{1}{2} kx^2 = \frac{1}{2} mv^2$$

$$Kx^2 = mv^2$$

$$v = x \sqrt{\frac{k}{m}} = 0.05 \sqrt{\frac{100}{0.1}} = 0.05 \times 10\sqrt{10}$$

$$v = 0.5\sqrt{10}$$

$$\text{From } h = \frac{1}{2} gt^2$$

$$t = \sqrt{\frac{2h}{g}} = \sqrt{\frac{2 \times 2}{10}} = \frac{2}{\sqrt{10}}$$

$$\therefore d = vt = 0.5\sqrt{10} \times \frac{2}{\sqrt{10}} = 1 \text{ m}$$

2. Ans (3)

$$\tan \phi = \frac{x_C - x_L}{R}$$

$$\tan 45 = \frac{x_C - x_L}{R}$$

$$x_C - x_L = R$$

$$\frac{1}{\omega C} - \omega L = R$$

$$\frac{1}{\omega C} - 300 \times 0.03 = 1$$

$$\frac{1}{\omega C} = 10$$

$$C = \frac{1}{10\omega} = \frac{1}{10 \times 300}$$

$$C = \frac{1}{3} \times 10^{-3}$$

$$X = 3$$

3. Ans (2)

$$\text{At } t = 0, y = \frac{1}{1+x^2}$$

$$\text{At time } t = t, y = \frac{1}{1+(x-vt)^2}$$

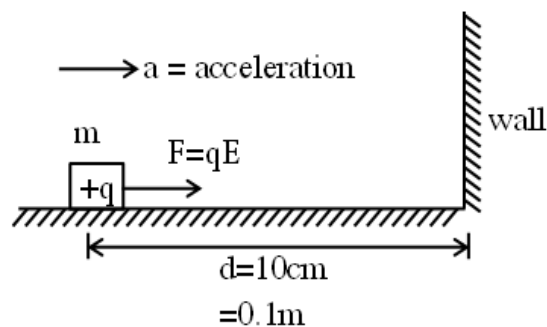
$$\text{At } t = 1, y = \frac{1}{1+(x-v)^2} \dots\dots(i)$$

$$\text{At } t = 1, y = \frac{1}{1+(x-2)^2} \dots\dots(ii)$$

Comparing (i) & (ii)

$$v = 2\text{m/s}$$

4. Ans (1)



$$F = ma$$

$$qE = ma$$

$$a = \frac{qE}{m}$$

$$\text{Now } d = \frac{1}{2}at^2$$

$$t = \sqrt{\frac{2d}{a}}$$

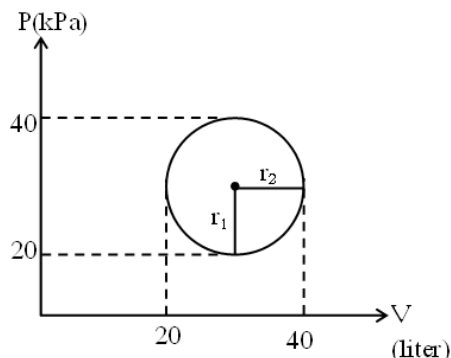
$$t = \sqrt{\frac{2d}{\left(\frac{qE}{m}\right)}}$$

$$t = \sqrt{\frac{2 \times 0.1}{\left(\frac{8 \times 10^{-6}}{10^{-3}}\right) \times 100}} = \frac{1}{2}$$

$$\therefore \text{Time period} = 2t = 1 \text{ sec}$$

$$\text{Ans.} = 1.00$$

5. Ans (100)



For complete cyclic process

$$\Delta U = 0$$

$$\therefore \text{from } \Delta Q = \Delta U + W$$

$$= 0 + W$$

$$\Delta Q = W$$

$$= \text{Area}$$

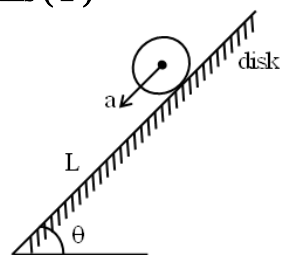
$$= \pi r_1 \cdot r_2$$

$$= \pi \times (10 \times 10^3) \times (10 \times 10^{-3})$$

$$\Delta Q = 100 \pi$$

$$\therefore \text{Ans.} = 100$$

6. Ans (2)



If disk slips on inclined plane, then its acceleration

$$a_1 = g \sin \theta$$

$$L = \frac{1}{2} a_1 t_1^2 \quad \Rightarrow t_1 = \sqrt{\frac{2L}{a_1}} \dots(i)$$

If disk rolls on inclined plane, its acceleration,

$$a_2 = \frac{g \sin \theta}{1 + \frac{1}{mR^2}}$$

$$a_2 = \frac{g \sin \theta}{1 + \frac{mR^2}{2mR^2}}$$

$$a_2 = \frac{2}{3} g \sin \theta$$

$$\text{Now } L = \frac{1}{2} a_2 \cdot t_2^2 \quad \Rightarrow t_2 = \sqrt{\frac{2L}{a_2}} \dots(ii)$$

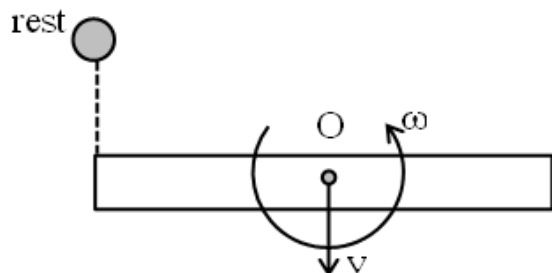
$$\text{Now } \frac{t_2}{t_1} = \sqrt{\frac{a_1}{a_2}} = \sqrt{\frac{3}{2}}$$

$$\Rightarrow x = 2$$

7. Ans (4)



Just before collision



Just after collision

From momentum conservation, $P_i^0 = P_f$

$$mu = Mv \quad \dots (i)$$

From angular momentum conservation about O,

$$mu \cdot \frac{L}{2} = \frac{ML^2}{12} \omega$$

$$\Rightarrow \omega = \frac{6mu}{ML} \quad \dots (ii)$$

$$\text{From } e = \frac{R \cdot V \cdot S}{R \cdot V \cdot A}$$

$$1 = \frac{V + \frac{\omega L}{2}}{u}$$

$$v + \frac{\omega L}{2} = u$$

$$v + \frac{3mu}{M} = u$$

$$\frac{mu}{M} + \frac{3mu}{M} = u$$

$$\frac{4mu}{M} = u$$

$$\frac{m}{M} = \frac{1}{4}$$

$$X = 4$$

8. Ans (25)

For simple microscope,

$$m = 1 + \frac{D}{f_0}$$

$$6 = 1 + \frac{D}{f_0}$$

$$5 = \frac{25}{f_0}$$

$$f_0 = 5 \text{ cm}$$

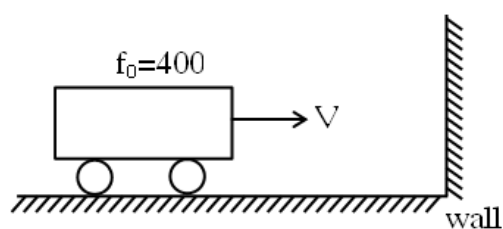
For compound microscope,

$$m = \frac{\ell \cdot D}{f_0 \cdot f_e}$$

$$12 = \frac{60 \times 25}{5 \cdot f_e}$$

$$f_e = 25 \text{ cm}$$

9. Ans (132)



Wall as an observer

Frequency received by wall

$$f_1 = f_0 \left(\frac{C}{C - V} \right)$$

Again wall as a source

Frequency received by observer on car

$$f_2 = f_1 \left(\frac{C + V}{C} \right)$$

$$f_2 = f_0 \left(\frac{C + V}{C - V} \right)$$

$$500 = 400 \left(\frac{C + V}{C - V} \right)$$

$$\frac{5}{4} = \frac{C + V}{C - V}$$

$$C = 9V$$

$$V = \frac{C}{9} = \frac{330}{9} \text{ m/s}$$

$$V = \frac{330}{9} \times \frac{18}{5} = 132 \text{ km/hr}$$

10. Ans (40)

Maximum amplitude

$$A_{\max} = A_m + A_c$$

$$\Rightarrow V_{\max} = V_m + V_c$$

$$200 = V_m + 160$$

$$V_m = 40$$

$\therefore$ Peak voltage $A_m = 40$

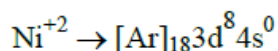
Ans. 40

PART-2 : CHEMISTRY

SECTION-I

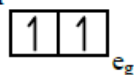
1. Ans (B)

1) $\text{NiCl}_2 \cdot 6\text{H}_2\text{O}$

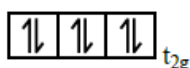


C.N. = 6 octahedral

splitting



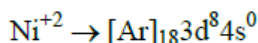
Hybridisation
 $sp^3 d^2$



t_{2g}

2) $\text{K}_2[\text{Ni}(\text{CN})_4]$

C.N. 4



$\text{CN}^- \rightarrow$ Strong field ligand



Hybridisation
 dsp^2

Square planar splitting



$d_{xy} \quad d_{xz}$

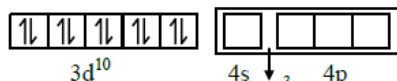
3) $\text{Ni}(\text{CO})_4$

CO -

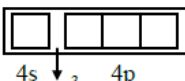
Strong

field ligand

$\text{Ni} \rightarrow [\text{Ar}]_{18}$

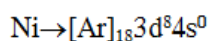


$3d^{10}$



Hybridisation
 sp^3

4) $\text{Na}_2[\text{NiCl}_4]$



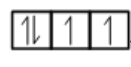
$\text{Cl}^- \rightarrow$ weak

field ligand

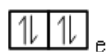
C.N. 4

tetrahedral

splitting

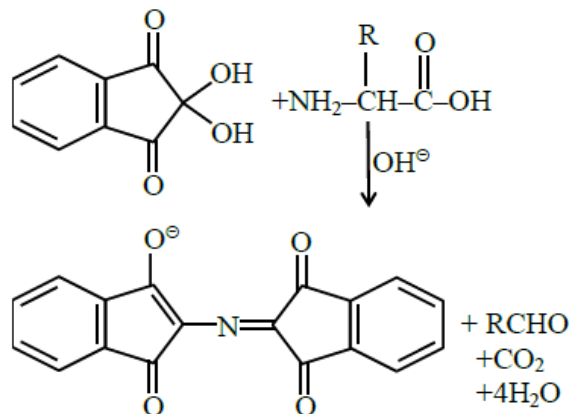


Hybridisation
 sp^3



e

2. Ans (D)



Ninhydrin Test

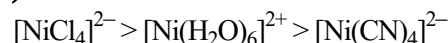
3. Ans (D)

Chlorine gas was used earlier for bleaching paper. These days, hydrogen peroxide (H_2O_2) with suitable catalyst.

Tetra chloroethene ($\text{Cl}_2\text{C}=\text{CCl}_2$) was earlier used as solvent for dry cleaning. The compound contaminates the ground water and is also a suspected carcinogen. Replacement of halogenated solvent by liquid CO_2 will result in less harm to groundwater.

Hence given statement (4) is correct.

4. Ans (C)



Splitting $\Delta_t < \Delta_0 < \Delta_{sq}$

energy order

absorbed $[\text{NiCl}_4]^{2-} < [\text{Ni}(\text{H}_2\text{O})_6]^{2+} < [\text{Ni}(\text{CN})_4]^{2-}$

energy order

intensity of $[\text{NiCl}_4]^{2-} > [\text{Ni}(\text{H}_2\text{O})_6]^{2+} > [\text{Ni}(\text{CN})_4]^{2-}$

colour of

compound

5. Ans (B)

1) $\text{B}(\text{OH})_3$ acidic and H_3PO_3 acidic

2) $\text{B}(\text{OH})_3$ acidic and $\text{Al}(\text{OH})_3$ amphoteric

3) NaOH basic and $\text{Ca}(\text{OH})_2$ basic

4) $\text{Be}(\text{OH})_2$ amphoteric and $\text{Al}(\text{OH})_3$ amphoteric

6. Ans (A)

In BrO_4^- , Br is in highest oxidation state (+7),

So it cannot oxidise further hence it cannot show disproportionation reaction.

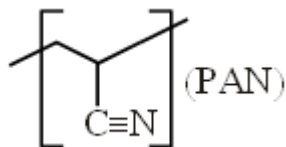
7. **Ans (B)**

Hence given statement (A) is not correct

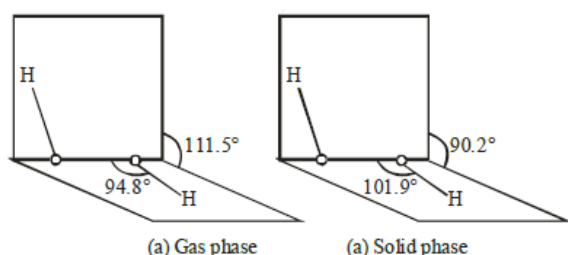
But statement (B) is correct

8. **Ans (A)**

→ orlon fibers are made up of Polyacrylonitrile



9. **Ans (D)**

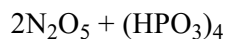
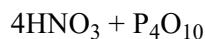


(a) H_2O_2 structure in gas phase, dihedral angle is 111.5° .

(b) H_2O_2 structure in solid phase at 110K, dihedral angle is 90.2° .

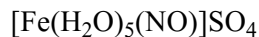
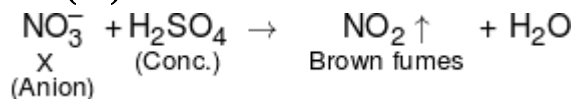
Hence given statement (A) is not correct But statement (B) is correct.

10. **Ans (A)**

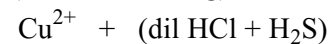


Ans. N_2O_5 is acidic in nature.

11. **Ans (C)**

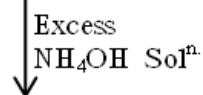
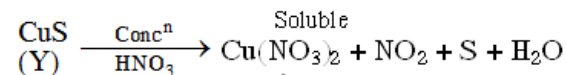


(Dark brown ring)

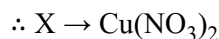


(Black ppt)

(Y)



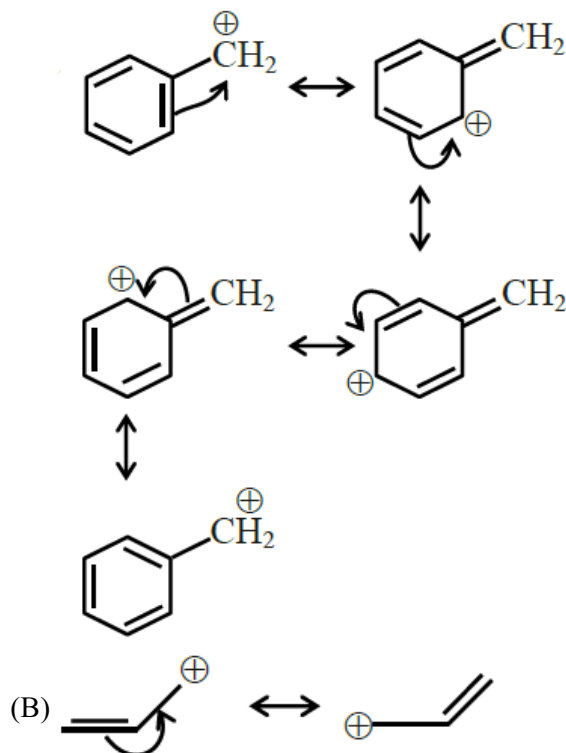
Deep blue colour solution.



12. **Ans (C)**

(A) and (B) only in Resonance

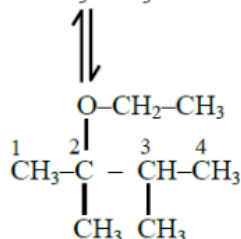
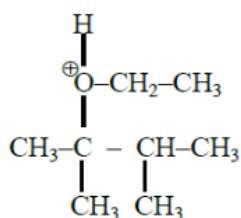
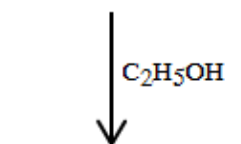
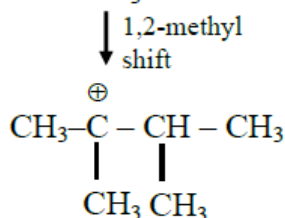
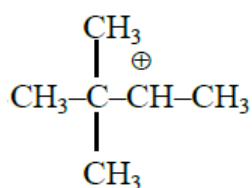
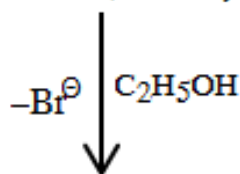
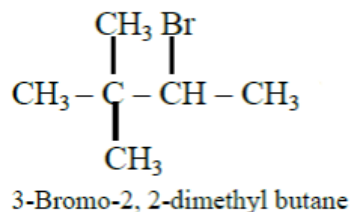
(A)



13. Ans (D)

- (A) $2\text{Mg} + \text{O}_2 \rightarrow 2\text{MgO}$ (Diamagnetic)
 (B) $2\text{Na} + \text{O}_2 \rightarrow \text{Na}_2\text{O}$ (Diamagnetic)
 $2\text{Na} + \text{O}_2 \xrightarrow{\text{excess}} \text{Na}_2\text{O}_2$ (Diamagnetic)
 (C) $2\text{Ca} + \text{O}_2 \rightarrow 2\text{CaO}$ (Diamagnetic)
 $\text{Ca} + \text{O}_2 \rightarrow \text{CaO}_2$ (Diamagnetic)
 (D) $\text{K} + \text{O}_2 \xrightarrow{\text{excess}} \text{KO}_2$ (Paramagnetic)

14. Ans (C)

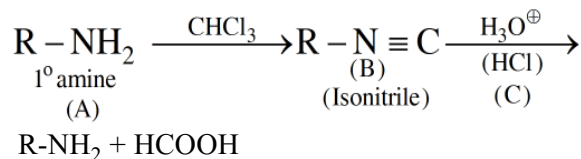


2 Ethoxy -2,3-dimethyl butane

15. Ans (B)

Zinc can be purified economically by fractional distillation.

16. Ans (C)



17. Ans (A)

The phenomenon of scattering of light by colloidal particles as a result of which the path of the beam becomes visible is called a tyndall effect.

smaller the diameter and similar the magnitude of refractive indices, lesser is the scattering and hence the tyndall effect and viced-versa.

The diameter of the dispersed phase particle should not be smaller than the wavelength of light used because they won't be able to scatter the light so, therefore, the diameter of the dispersed particles should be equal or not much smaller than the wavelength of the light used.

2. The refractive indices (i.e. the ratio of the velocity of light in vacuum to the velocity of light in any medium) of the dispersed phase and the dispersion medium should differ greatly in magnitude than only the particles will be able to scatter the light and tyndall effect will be observed. On the other hand, if the refractive indices of the dispersed phase and dispersion medium are almost similar in magnitude, then there will be no scattering of light and hence, therefore, no tyndall effect is observed.

Hence answer A and E are correct.

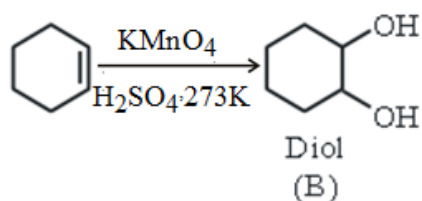
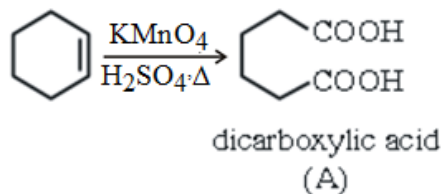
18. Ans (A)

Amylose is a linear chain polymer of α -D-glucose while amylopectine is branched chain polymer of α -D-glucose.

19. Ans (C)

	Aldehydes give $\oplus$ ve Tollen's Test (Silver mirror test)	Tollen's test
(I)		Positive
(II)		Negative
(III)		Positive
(IV)		Positive

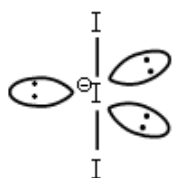
20. Ans (D)



PART-2 : CHEMISTRY

SECTION-II

1. Ans (3)



The number of lone pairs of electron on the central atom is 3.

2. Ans (226)

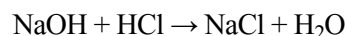
We known that no. of moles = $V_{\text{litre}} \times \text{Molarity}$

& No. of millimoles = $V_{\text{ml}} \times \text{Molarity}$

so millimoles of NaOH = $250 \times 0.5 = 12.5$

Millimoles of HCl = $500 \times 1 = 500$

Now reaction is



t = 0 125 500 0 0

t = t 0 375 125 125

so millimoles of HCl left = 375

Moles of HCl = 375×10^{-3}

No. of HCl molecules = $6.022 \times 10^{23} \times 375 \times 10^{-3}$

= $225.8 \times 10^{21} \approx 226 \times 10^{21} = 226$

3. Ans (0)

$\text{Ga}^+ : 1s^2 2s^2 2p^6 3s^2 3p^6 3d^{10} 4s^2$

The azimuthal quantum number for the valence electrons (4s-subshell) of Ga^+ ion is zero(0).

4. Ans (2)

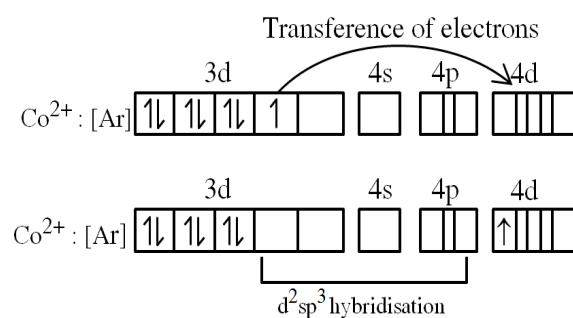
$[\text{Co}(\text{CN})_6]^{4-}$

$x + 6 \times (-1) = -4$

$x = +2$

$\text{Co}^{2+} : [\text{Ar}] 3d^7$

and CN^- is a strong field ligand which can pair electron of central atom.



It has one unpaired electron (n) in 4d-subshell.

So spin only magnetic moment

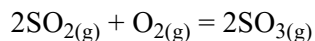
$$(\mu) = \sqrt{n(n+2)} \text{ B. M}$$

where n = number of unpaired electrons.

$$\mu = \sqrt{3} \text{ B. M}$$

$$\mu = 1.73 \text{ BM}$$

5. Ans (172)



$$K_P = \frac{(p\text{SO}_{3(g)})^2}{p\text{SO}_{2(g)} \times p\text{O}_{2(g)}}$$

$$= \frac{43 \times 43}{45 \times 45} \times 530 \text{ Pa}^{-1}$$

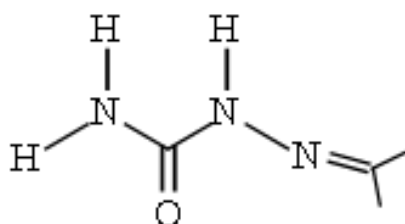
$$= 172.28 \times 10^{-5} \text{ Pa}^{-1}$$

$$= 172.28 \text{ atm}$$

$$= 17228 \times 10^{-2} \text{ atm}$$

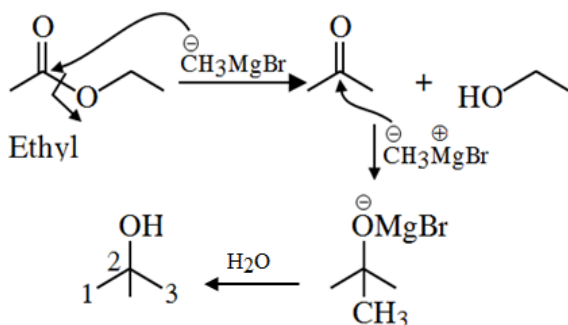
Ans is 17228

6. Ans (3)



Semicarbazone molecule of acetone

7. Ans (2)



2-Methylpropan-2-ol

8. Ans (106)

As the unit of rate constant is min^{-1} so it must be a first order reaction

$$K \times t = 2.303 \log A_0/A_t$$

in 1 min 10% is in activated so tabing

$$A_0 = 100 \quad A_t = 90 \text{ in 1 min}$$

$$\text{So } K \times 1 = 2.303 \times \log \frac{100}{90}$$

$$= 2.303 \times (\log 10 - 2\log 3)$$

$$= 2.303 \times (1 - 2 \times 0.477)$$

$$= 0.10593$$

$$= 105.93 \times 10^{-3} \approx 106$$

9. Ans (667)

1 mole glucose give 2700 kJ energy

so mole of glucose needed for 10^5 kJ energy

$$= \frac{10000}{2700} = 370 \text{ moles}$$

$$\text{wt. of glucose} = 3.10 \times 180 = 666.666$$

$$\approx 667 \text{ gm}$$

$$\frac{Y_{\text{Benzene}}}{Y_{\text{M.B}}} = \frac{P_B^0 X_B}{P_{\text{MB}}^0 X_{\text{MB}}} = \frac{70 \times 1}{20 \times 1} = \frac{7}{2}$$

$$Y_{\text{Benzene}} = \frac{7}{9} = 77.77 \times 10^{-2}$$

$$= 78 \times 10^{-12}$$

10. Ans (78)

$$P_B^0 = 40 \quad P_T^0 = 20 \quad K_B = 0.5 = K_T$$

$$\text{Now } y_B = \frac{K_B P_B^0}{K_B P_B^0 + K_T P_T^0} = \frac{70 \times 0.5}{70 \times 0.5 + 20 \times 0.5}$$

PART-3 : MATHEMATICS

SECTION-I

1. Ans (B)

p	q	$\sim p$	$\sim q$	$p \wedge q$	$q \vee \sim p$	$(p \wedge \sim q) \Rightarrow (q \vee \sim p)$	$p \Rightarrow q$
T	F	F	T	T	F	F	F
F	T	T	F	F	T	T	T
T	T	F	F	F	T	T	T
F	F	T	T	F	T	T	T

$$\therefore (p \wedge \sim q) \Rightarrow (q \vee \sim p)$$

$$\equiv p \Rightarrow q$$

2. **Ans (B)**

$$a > 0$$

$$\text{Let } n \leq a < n+1, n \in \mathbb{W}$$

$$\therefore a = [a] + \{a\}$$

$$\Downarrow \quad \Downarrow$$

G.I.F Fractional part

$$\text{Here } [a] = n$$

$$\text{Now, } \int_0^a e^{x-[x]} dx = 10e - 9$$

$$\Rightarrow \int_0^n e^{\{x\}} dx + \int_n^a e^{x-[x]} dx = 10e - 9$$

$$\therefore n \int_0^1 e^x dx + \int_n^a e^{x-n} dx = 10e - 9$$

$$\Rightarrow n(e - 1) + (e^{a-n} - 1) = 10e - 9$$

$$\therefore \boxed{n = 0} \text{ and } \{a\} = \log_e 2$$

$$\text{So, } a = [a] + \{a\} = (10 + \log_e 2)$$

3. **Ans (A)**

$$\text{Let other two numbers be } a, (21 - a)$$

Now,

$$10.25 = \frac{(4 + 16 + 25 + 49 + a^2 + (21 - a)^2)}{6} - (6.5)^2$$

(Using formula for variance)

$$\Rightarrow 6(10.25) + 6(6.5)^2 = 94 + a^2 + (21 - a)^2$$

$$\Rightarrow a^2 + (21 - a)^2 = 221$$

$$\therefore a = 10 \text{ and } (21 - a) = 21 - 10 = 11$$

So, remaining two observations are 10, 11.

4. **Ans (C)**

$$\text{Let } I = 2 \int_0^1 \underbrace{\ln(\sqrt{1-x} + \sqrt{1+x})}_{(I)} \underbrace{1}_{(II)} dx$$

(I. B. P.)

$$\therefore I = 2 \left[(x \cdot \ln(\sqrt{1-x} + \sqrt{1+x})) \Big|_0^1 - \int_0^1 x \cdot \left(\frac{1}{\sqrt{1-x} + \sqrt{1+x}} \right) \cdot \left(\frac{1}{2\sqrt{1+x}} - \frac{1}{2\sqrt{1-x}} \right) dx \right]$$

$$= 2(\ln \sqrt{2} - 0) - \frac{2}{2} \int_0^1 \frac{x\sqrt{1-x} - \sqrt{1+x} dx}{(\sqrt{1-x} + \sqrt{1+x})\sqrt{1-x^2}}$$

$$= (\log_e 2) - \int_0^1 x \cdot \frac{(2 - 2\sqrt{1-x^2})}{-2x\sqrt{1-x^2}} dx$$

(After rationalisation)

$$= (\log_e 2) + \int_0^1 \left(\frac{1 - \sqrt{1-x^2}}{\sqrt{1-x^2}} \right) dx$$

$$= (\log_e 2) + (\sin^{-1} x) \Big|_0^1 - 1$$

$$= \log_e 2 + \left(\frac{\pi}{2} - 0 \right) - 1$$

$$\therefore I = (\log_e 2) + \frac{\pi}{2} - 1$$

5. **Ans (C)**

$$\text{As, } (\alpha^2 + \sqrt{3}) = -(3)^{1/4} \cdot \alpha$$

$$\Rightarrow (\alpha^4 + 2\sqrt{3}\alpha^2 + 3) = \sqrt{3}\alpha^2 \text{ (On squaring)}$$

$$\therefore (\alpha^4 + 3) = (-)\sqrt{3}\alpha^2$$

$$\Rightarrow \alpha^8 + 6\alpha^4 + 9 = 3\alpha^4 \text{ (Again squaring)}$$

$$\therefore \alpha^8 + 3\alpha^4 + 9 = 0$$

$$\Rightarrow \boxed{\alpha^8 = -9 - 3\alpha^4}$$

(Multiply by α^4)

$$\text{So, } \alpha^{12} = -9\alpha^4 - 3\alpha^8$$

$$\therefore \alpha^{12} = -9\alpha^4 - 3(-9 - 3\alpha^4)$$

$$\Rightarrow \alpha^{12} = -9\alpha^4 + 27 + 9\alpha^4$$

$$\text{Hence, } \boxed{\alpha^{12} = (27)^2}$$

$$\Rightarrow (\alpha^{12})^8 = (27)^8$$

$$\Rightarrow \alpha^{96} = (3)^{24}$$

$$\text{Similarly } \beta^{96} = (3)^{24}$$

$$\therefore \alpha^{96}(\alpha^{12} - 1) + \beta^{96}(\beta^{12} - 1) = (3)^{24} \times 52$$

6. **Ans (A)**

$$A = \begin{bmatrix} 2 & 3 \\ a & 0 \end{bmatrix}, a \in \mathbb{R}$$

$$\text{and } P = \frac{A + A^T}{2} = \begin{bmatrix} 2 & \frac{3+a}{2} \\ \frac{a+3}{2} & 0 \end{bmatrix}$$

$$\text{and } Q = \frac{A - A^T}{2} = \begin{bmatrix} 0 & \frac{3-a}{2} \\ \frac{a-3}{2} & 0 \end{bmatrix}$$

$$\text{As, } \det(Q) = 9$$

$$\Rightarrow (a-3)^2 = 36$$

$$\Rightarrow a = 3 \pm 6$$

$$\therefore a = 9, -3$$

$$\therefore \det(P) = \begin{vmatrix} 2 & \frac{3+a}{2} \\ \frac{a+3}{2} & 0 \end{vmatrix}$$

$$= 0 - \frac{(a-3)^2}{4} = 0, \text{ for } a = -3$$

$$= 0 - \frac{(a-3)^2}{4} = -\frac{1}{4}(12)(12), \text{ for } a = 9$$

$$\therefore \text{Modulus of the sum of all possible values of } \det(P) = |-36| + |0| = 36 \text{ Ans.}$$

7. **Ans (B)**

$$\text{As } |z\omega| = 1$$

$$\Rightarrow \text{If } |z| = r, \text{ then } |\omega| = \frac{1}{r}$$

$$\text{Let } \arg(z) = \theta$$

$$\therefore \arg(\omega) = \left(\theta - \frac{3\pi}{2}\right)$$

$$\text{So, } z = re^{i\theta} \Rightarrow \bar{z} = re^{i(-\theta)}$$

$$\omega = \frac{1}{r}e^{i\left(\theta - \frac{3\pi}{2}\right)}$$

$$\text{Now, consider}$$

$$\frac{1 - 2\bar{z}\omega}{1 + 3\bar{z}\omega} = \frac{1 - 2e^{i\left(\theta - \frac{3\pi}{2}\right)}}{1 + 3e^{i\left(\theta - \frac{3\pi}{2}\right)}} = \left(\frac{1 - 2i}{1 + 3i}\right)$$

$$= \frac{(1 - 2i)(1 - 3i)}{(1 + 3i)(1 - 3i)} = -\frac{1}{2}(1 + i)$$

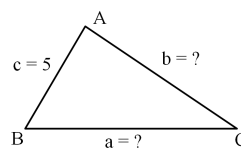
$$\therefore \text{prin arg}\left(\frac{1 - 2\bar{z}\omega}{1 + 3\bar{z}\omega}\right)$$

$$= \text{prin arg}\left(\frac{1 - 2\bar{z}\omega}{1 + 3\bar{z}\omega}\right)$$

$$= \left(-\frac{1}{2}(1 + i)\right)$$

$$= -\left(\pi - \frac{\pi}{4}\right) = -\frac{3\pi}{4}$$

8. **Ans (C)**



$$\text{As, } \cos B = \frac{3}{5} \Rightarrow B = 53^\circ$$

$$\text{As, } R = 5 \Rightarrow \frac{c}{\sin C} = 2R$$

$$\Rightarrow \frac{5}{10} = \sin C \Rightarrow C = 30^\circ$$

$$\text{Now, } \frac{b}{\sin B} = 2R \Rightarrow b = 2(5)\left(\frac{4}{5}\right) = 8$$

Now, by cosine formula

$$\cos B = \frac{a^2 + c^2 - b^2}{2ac} \Rightarrow \frac{3}{5} = \frac{a^2 + 25 - 64}{2(5)a}$$

$$\Rightarrow a^2 - 6a - 39 = 0$$

$$\therefore a = \frac{6 \pm \sqrt{192}}{2} = \frac{6 \pm 8\sqrt{3}}{2}$$

$$\Rightarrow 3 + 4\sqrt{3} \text{ (Reject } a = 3 - 4\sqrt{3})$$

$$\text{Now, } \Delta = \frac{abc}{4R}$$

$$= \frac{(3 + 4\sqrt{3})(8)(5)}{4(5)} = 2(3 + 4\sqrt{3})$$

$$\Rightarrow \Delta = (6 + 8\sqrt{3})$$

9. **Ans (C)**

For domain,

$$\frac{||x|| - 2}{||x|| - 3} \geq 0$$

$$\text{Case I : When } ||x|| - 2 \geq 0$$

$$\text{and } ||x|| - 3 > 0$$

$$\therefore x \in (-\infty, -3) \cup [4, \infty) \dots (1)$$

$$\text{Case II : When } ||x|| - 2 \leq 0$$

$$\text{and } ||x|| - 3 < 0$$

$$\therefore x \in [-2, 3) \dots (2)$$

So, from (1) and (2)

we get

Domain of function

$$= (-\infty, -3) \cup [-2, 3) \cup [4, \infty)$$

$$\therefore (a + b + c) = -3 + (-2) + 3 = -2 \text{ (} a < b < c \text{)}$$

10. Ans (A)

We have

$$\frac{dy}{dx} = \frac{x \left(\frac{y}{x} \cdot \tan \frac{y}{x} - 1 \right)}{x \tan \frac{y}{x}}$$

$$\therefore \frac{dy}{dx} = \frac{y}{x} - \cot\left(\frac{y}{x}\right)$$

Put $\frac{y}{x} = v$

$$\Rightarrow y = vx$$

$$\therefore \frac{dy}{dx} = v + \frac{ndv}{dx}$$

Now, we get

$$v + n \frac{dv}{dx} = v - \cot(v)$$

$$\Rightarrow \int (\tan v) dv = - \int \frac{dx}{x}$$

$$\therefore \ln \left| \sec\left(\frac{y}{x}\right) \right| = -\ln |x| + c$$

$$\text{As } \left(\frac{1}{2}\right) = \left(\frac{y}{x}\right) \Rightarrow \boxed{C=0}$$

$$\therefore \sec\left(\frac{y}{x}\right) = \frac{1}{x}$$

$$\Rightarrow \cos\left(\frac{y}{x}\right) = x$$

$$\therefore \boxed{y = x \cos^{-1}(x)}$$

So, required bounded area

$$= \int_0^{\frac{1}{\sqrt{2}}} \underset{(II)}{x} \underset{(I)}{(\cos^{-1} x)} dx = \left(\frac{\pi-1}{8} \right)$$

(I.B.P.)

11. Ans (B)

$$(1-x)^{100} \cdot (x^2+x+1)^{100} \cdot (1-x)$$

$$= ((1-x)(x^2+x+1))^{100} (1-x)$$

$$= (1^3 - x^3)^{100} (1-x)$$

$$= (1-x^3)^{100} (1-x)$$

$$= \underbrace{(1-x^3)^{100}}_{\text{No term of } x^{256}} - \underbrace{x(1-x^3)^{100}}_{\text{We find coefficient of } x^{255}}$$

$$\text{Required coefficient } (-1) \times (-^{100}C_{85})$$

$$= {}^{100}C_{85} = {}^{100}C_{15}$$

12. Ans (D)

$$A = \begin{bmatrix} 1 & -x & 2x+1 \\ -x & 1 & -x \\ 2x+1 & -x & 1 \end{bmatrix}$$

$$|A| = 4x^3 - 4x^2 - 4x = f(x)$$

$$f'(x) = 4(3x^2 - 2x - 1) = 0$$

$$\Rightarrow x = 1 ; x = -\frac{1}{3}$$

$$\therefore \underbrace{f(1) = -4}_{\min} ; \underbrace{f\left(-\frac{1}{3}\right) = \frac{20}{27}}_{\max.}$$

$$\text{Sum} = -4 + \frac{20}{27} = -\frac{88}{27}$$

13. Ans (D)

$$|\vec{a}| = 3 = a ; \vec{a} \cdot \vec{c} = c$$

$$\text{Now } |\vec{c} - \vec{a}| = 2\sqrt{2}$$

$$\Rightarrow c^2 + a^2 - 2\vec{c} \cdot \vec{a} = 8$$

$$\Rightarrow c^2 + 9 - 2(c) = 8$$

$$\Rightarrow c^2 - 2c + 1 = 0 \Rightarrow c = 1 = |\vec{c}|$$

$$\text{Also, } \vec{a} \times \vec{b} = 2\hat{i} - 2\hat{j} + \hat{k}$$

$$\text{Given } (\vec{a} \times \vec{b}) = |\vec{a} \times \vec{b}| |\vec{c}| \sin \frac{\pi}{6}$$

$$= (3)(1)(1/2)$$

$$= 3/2$$

14. Ans (D)

$$\tan^{-1} \sqrt{x^2+x} + \sin^{-1} \sqrt{x^2+x+1} = \frac{\pi}{4}$$

For equation to be defined,

$$x^2 + x \geq 0$$

$$\Rightarrow x^2 + x + 1 \geq 1$$

$\therefore$ only possibility that the equation is defined

$$x^2 + x = 0 \Rightarrow x = 0; x = -1$$

None of these values satisfy

$\therefore$ No of roots = 0

15. **Ans (B)**

$$e^x \sqrt{1-y^2} dx + \frac{y}{x} dy = 0$$

$$\Rightarrow e^x \sqrt{1-y^2} dx + \frac{-y}{x} dy$$

$$\Rightarrow \int \frac{-y}{\sqrt{1-y^2}} dy = \int \frac{e^x}{x} dx$$

$$\Rightarrow \sqrt{1-y^2} = e^x (x-1) + c$$

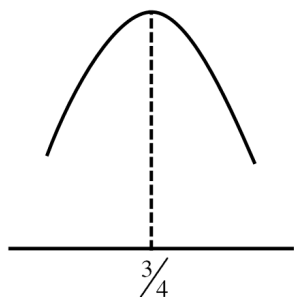
$$\text{Given : At } x=1, y=-1$$

$$\Rightarrow 0 = 0 + c \Rightarrow c = 0$$

$$\therefore \sqrt{1-y^2} = e^x (x-1)$$

$$\text{At } x=3 \quad 1-y^2 = (e^3 \cdot 2)^2 \Rightarrow y^2 = 1 - 4e^6$$

16. **Ans (A)**



$$\frac{-B}{2A} = \frac{3}{4}$$

$$\Rightarrow \frac{-(6)}{2a} = \frac{3}{4}$$

$$\Rightarrow a = \frac{-6 \times 4}{6} \Rightarrow a = -4$$

$$\therefore g(x) = 4x^2 - 6x + 15$$

$$\text{Local max. at } x = \frac{-B}{2A} = -\frac{(-6)}{2(-4)} = \frac{-3}{4}$$

17. **Ans (B)**

Continuous at $x=0$

$$f(0^+) = f(0^-) \Rightarrow a - 1 = 0 - e^0$$

$$\Rightarrow a = 0$$

Continuous at $x=1$

$$f(1^+) = f(1^-)$$

$$\Rightarrow 2(1) - b = a + (-1)$$

$$\Rightarrow b = 2 - a + 1 \Rightarrow b = 3$$

$$\therefore a + b = 3$$

18. **Ans (B)**

AAEIIIMNNOTX

-----M-----

$$\text{Total words with M at fourth Place} = \frac{10!}{2!2!2!}$$

$$\text{Total words} = \frac{11!}{2!2!2!}$$

$$\text{Required probability} = \frac{10!}{11!} = \frac{1}{11}$$

19. **Ans (B)**

$$D < 0$$

$$\Rightarrow 4(a+4)^2 - 4(-5a+64) < 0$$

$$\Rightarrow a^2 + 16 + 8a + 5a - 64 < 0$$

$$\Rightarrow a^2 + 13a - 48 < 0$$

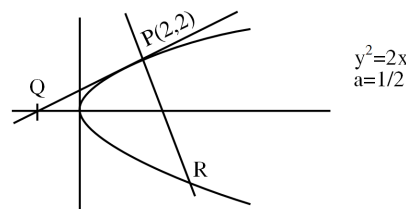
$$\Rightarrow (a+16)(a-3) < 0$$

$$\Rightarrow a \in (-16, 3)$$

$$\therefore \text{Possible } a : \{-5, -4, \dots, 3\}$$

$$\therefore \text{Required probability} = \frac{8}{36} = \frac{2}{9}$$

20. **Ans (A)**



$$\text{Tangent at P : } y(2) = 2 \left(\frac{1}{2} \right) (x + 2)$$

$$\Rightarrow 2y = x + 2$$

$$\therefore Q = (-2, 0)$$

$$\text{Normal at P : } y - 2 = -\frac{(2)}{2 \cdot \frac{1}{2}} (x - 2)$$

$$\Rightarrow y - 2 = -2(x - 2)$$

$$\Rightarrow y = 6 - 2x$$

$$\therefore \text{Solving with } y^2 = 2x \Rightarrow R \left(\frac{9}{2}, -3 \right)$$

$$\therefore \text{Ar}(\Delta PQR) = \frac{1}{2} \begin{vmatrix} 2 & 2 & 1 \\ -2 & 0 & 1 \\ \frac{9}{2} & -3 & 1 \end{vmatrix}$$

$$= \frac{25}{2} \text{ sq. units}$$

PART-3 : MATHEMATICS

SECTION-II

1. **Ans (4)**

$$|\vec{a} + \vec{b} + \vec{c}|^2 = |\vec{a}|^2 + |\vec{b}|^2 + |\vec{c}|^2 + 2(\vec{a} \cdot \vec{b} + \vec{a} \cdot \vec{c} + \vec{b} \cdot \vec{c}) = 3$$

$$\Rightarrow |\vec{a} + \vec{b} + \vec{c}| = \sqrt{3}$$

$$\vec{a} \cdot (\vec{a} + \vec{b} + \vec{c}) = |\vec{a}| + |\vec{a} + \vec{b} + \vec{c}| \cos \theta$$

$$\Rightarrow 1 = \sqrt{3} \cos \theta$$

$$\Rightarrow \cos 2\theta = -\frac{1}{3}$$

$$\Rightarrow 36 \cos^2 2\theta = \boxed{4}$$

2. **Ans (910)**

$$\text{Let } A = \begin{pmatrix} 1 & -1 & 0 \\ 0 & 1 & -1 \\ 0 & 0 & 1 \end{pmatrix} = I + C$$

where

$$I = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}, C = \begin{pmatrix} 0 & -1 & 0 \\ 0 & 0 & -1 \\ 0 & 0 & 0 \end{pmatrix}$$

$$C^2 = \begin{pmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix},$$

$$C^3 = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} = C^4 = C^5 = \dots$$

$$B = 7A^{20} - 20A^7 + 2I$$

$$= 7(I + C)^{20} - 20(I + C)^7 + 2I$$

$$= 7(I + 20C + {}^{20}C_2 C^2) - 20(I + 7C + {}^7C_2 C^2) + 2I$$

So

$$b_{13} = 7 \times {}^{20}C_2 - 20 \times {}^7C_2 = \boxed{910}$$

3. **Ans (81)**

Equation of plane :

$$\begin{vmatrix} x-1 & y-0 & z-1 \\ 1-1 & 2 & 1-1 \\ 1-0 & 0-1 & 1+2 \end{vmatrix} = 0$$

$$\Rightarrow 3x - z - 2 = 0$$

$$\vec{a} = \alpha \hat{i} + \beta \hat{j} + \gamma \hat{k} \parallel \text{to } 3x - z - 2 = 0$$

$$\Rightarrow \boxed{3\alpha - 8 = 0} \quad \dots (1)$$

$$\vec{a} \perp \hat{i} + 2\hat{j} + 3\hat{k}$$

$$\Rightarrow \alpha + 2\beta + 3\gamma = 0 \quad \dots (2)$$

$$\vec{a} \cdot (\hat{i} + \hat{j} + 2\hat{k}) = 0$$

$$\Rightarrow \alpha + \beta + 2\gamma = 0 \quad \dots (3)$$

on solving 1, 2 & 3

$$\alpha = 1, \beta = -5, \gamma = 3$$

$$\text{So } (\alpha - \beta + \gamma) = \boxed{81}$$

4. **Ans (21)**

$$\left(4^{\frac{1}{4}} + 5^{\frac{1}{6}}\right)^{120}$$

$$T_{r+1} = {}^{120}C_r (2^{1/2})^{120-r} (5)^{r/6}$$

$$\text{for rational terms } r = 6\lambda \quad 0 \leq r \leq 120$$

so total no of terms are 21.

5. **Ans (6)**

$$\text{If } \vec{r} = \vec{a} + \lambda \vec{b} \text{ and } \vec{r} = \vec{c} + \mu \vec{d}$$

then shortest distance between two lines is

$$L = \frac{(\vec{a} - \vec{c}) \cdot (\vec{b} \times \vec{d})}{|\vec{b} \times \vec{d}|}$$

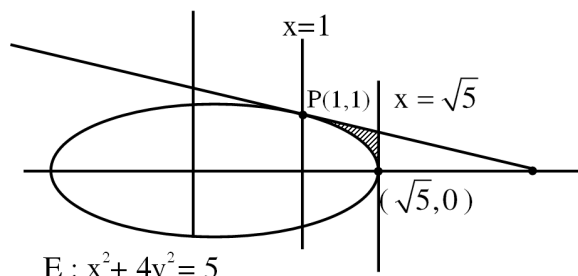
$$\therefore \vec{a} - \vec{c} = ((\alpha + 4)\hat{i} + 2\hat{j} + 3\hat{k})$$

$$\frac{\vec{b} \times \vec{d}}{|\vec{b} \times \vec{d}|} = \frac{(2\hat{i} + 2\hat{j} + \hat{k})}{3}$$

$$\therefore ((\alpha + 4)\hat{i} + 2\hat{j} + 3\hat{k}) \cdot \frac{(2\hat{i} + 2\hat{j} + \hat{k})}{3} = 9$$

$$\text{or } \alpha = 6$$

6. Ans (1)



$$E : x^2 + 4y^2 = 5$$

Tangent at P : $x + 4y = 5$

$$\text{Required Area} = \int_1^{\sqrt{5}} \left(\frac{5-x}{4} - \frac{\sqrt{5-x^2}}{2} \right) dx$$

$$= \left[\frac{5x}{4} - \frac{x^2}{8} - \frac{x}{4} \sqrt{5-x^2} - \frac{5}{2} \sin^{-1} \frac{x}{\sqrt{5}} \right]_1^{\sqrt{5}}$$

$$= \frac{5}{4} \sqrt{5} - \frac{5}{4} - \frac{5}{4} \cos^{-1} \left(\frac{1}{\sqrt{5}} \right)$$

It we assume $\alpha, \beta, \gamma, \in \mathbb{Q}$ (Not given in question)

$$\text{then } \alpha = \frac{5}{4}, \beta = -\frac{5}{4} \text{ \& } \gamma = -\frac{5}{4}$$

$$|\alpha + \beta + \gamma| = 1.25$$

7. Ans (1)

$$\begin{vmatrix} x+a-c & x+b & x+a \\ x-1 & x+c & x+b \\ x-b+d & x+d & x+c \end{vmatrix} = 2$$

$$C_2 \rightarrow C_2 - C_3$$

$$\Rightarrow \begin{vmatrix} x-2\lambda & \lambda & x+a \\ x-1 & \lambda & x+b \\ x+2\lambda & \lambda & x+c \end{vmatrix} = 2$$

$$R_2 \rightarrow R_2 - R_1, \quad R_3 \rightarrow R_3 - R_1$$

$$\Rightarrow \lambda \begin{vmatrix} x-2\lambda & 1 & x+a \\ 2\lambda-1 & 0 & \lambda \\ 4\lambda & 0 & 2\lambda \end{vmatrix} = 2$$

$$\Rightarrow 1(4\lambda^2 - 4\lambda^2 + 2\lambda) = 2$$

$$\Rightarrow \boxed{\lambda^2 = 1}$$

8. Ans (777)

15 : Players

6 : Bowlers

7 : Batsman

2 : Wicket keepers

Total number of ways for :

$$\text{at least 4 bowlers, 5 batsman \& 1 wicket keeper}$$

$$= {}^6C_4 ({}^7C_6 \times {}^2C_1 + {}^7C_5 \times {}^2C_2) + {}^6C_5 \times {}^7C_5 \times {}^2C_1$$

$$= \boxed{777}$$

9. Ans (34)

$$y^2 = -64x$$

focus : $(-16, 0)$

$y = mx + c$ is focal chord

$$\Rightarrow c = 16m \quad \dots\dots(1)$$

$y = mx + c$ is tangent to $(x+10)^2 + y^2 = 4$

$$\Rightarrow y = m(x+10) \pm 2\sqrt{1+m^2}$$

$$\Rightarrow c = 10m \pm 2\sqrt{1+m^2}$$

$$\Rightarrow 16m = 10m \pm 2\sqrt{1+m^2}$$

$$\Rightarrow 6m = 2\sqrt{1+m^2} \quad (m > 0)$$

$$\Rightarrow 9m^2 = 1 + m^2$$

$$\Rightarrow m = \frac{1}{2\sqrt{2}} \text{ \& } c = \frac{8}{\sqrt{2}}$$

$$4\sqrt{2}(m+c) = 4\sqrt{2} \left(\frac{17}{2\sqrt{2}} \right) = \boxed{34}$$

10. Ans (3)

$$\lim_{x \rightarrow 0} (2 - \cos x \sqrt{\cos x})^{\frac{x+2}{x^2}}$$

form : 1^∞

$$= e^{\lim_{x \rightarrow 0} \left(\frac{1 - \cos x \sqrt{\cos 2x}}{x^2} \right) \times (x+2)}$$

$$\text{Now } \lim_{x \rightarrow 0} \frac{1 - \cos x \sqrt{\cos 2x}}{x^2}$$

$$\lim_{x \rightarrow 0} \frac{\sin x \sqrt{\cos 2x} - \cos x \times \frac{1}{2\sqrt{\cos 2x}} \times (-2 \sin 2x)}{2x}$$

(by L' Hospital Rule)

$$\lim_{x \rightarrow 0} \frac{\sin x \cos 2x + \sin 2x \cdot \cos x}{2x}$$

$$= \frac{1}{2} + 1 = \frac{3}{2}$$

$$\lim_{x \rightarrow 0} \left(\frac{1 - \cos x \sqrt{\cos 2x}}{x^2} \right) (x+2)$$

So, $e^{\frac{3}{2} \times 2} = e^3$

$$= e^{\frac{3}{2} \times 2} = e^3$$

$$\Rightarrow \boxed{a = 3}$$